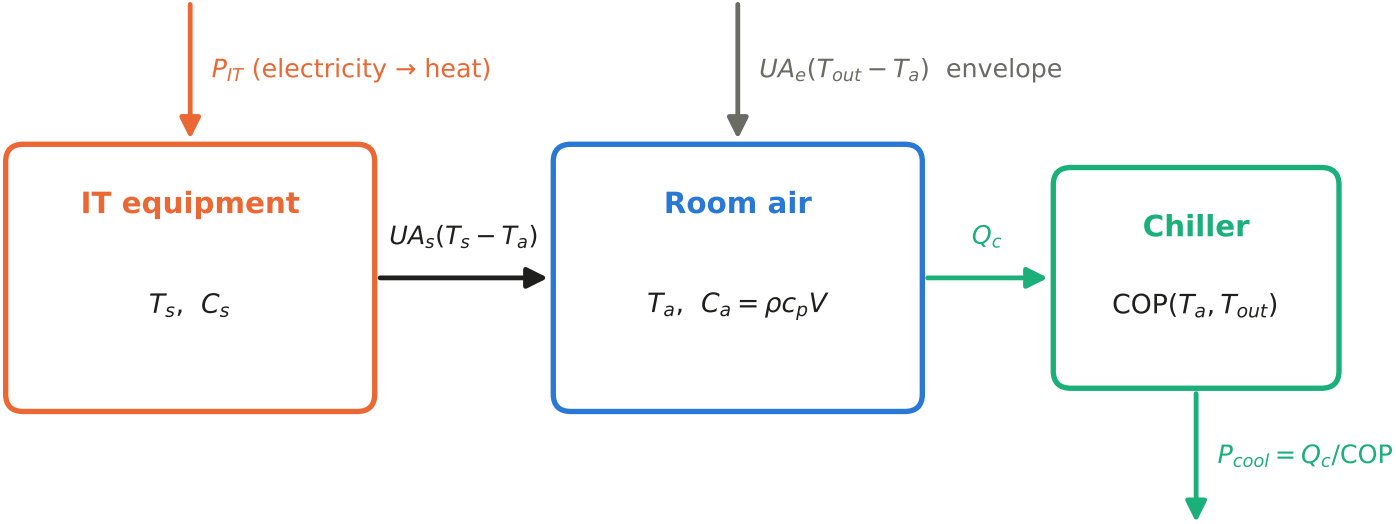

controller sets $Q_c \in [0, Q_{max}]$ every Δt (held constant: ZOH)